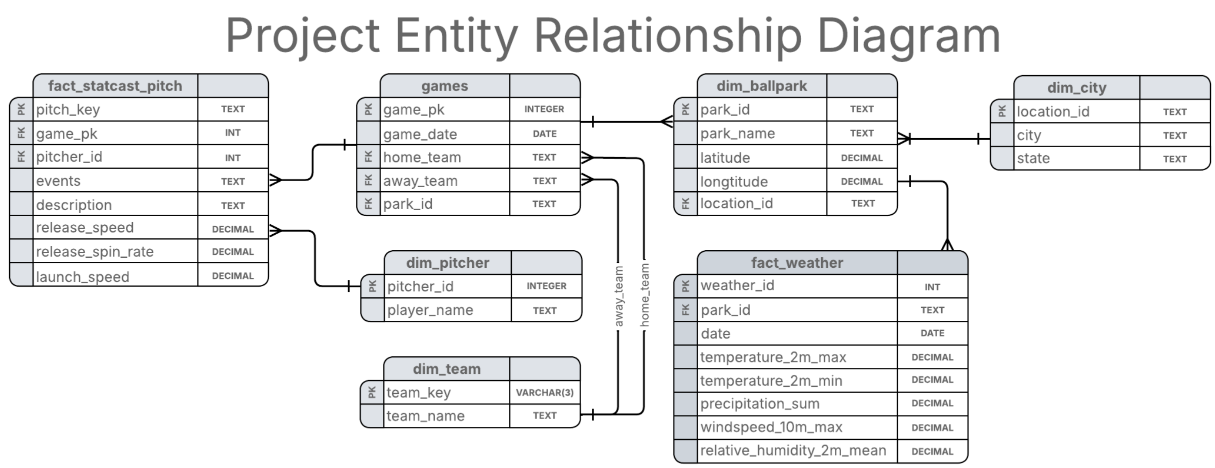
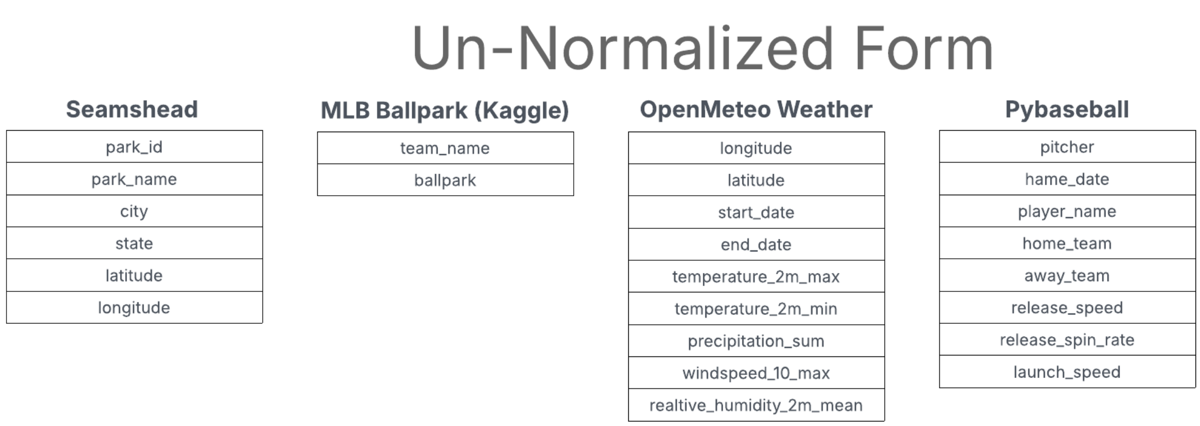
Entity Relationship Diagram

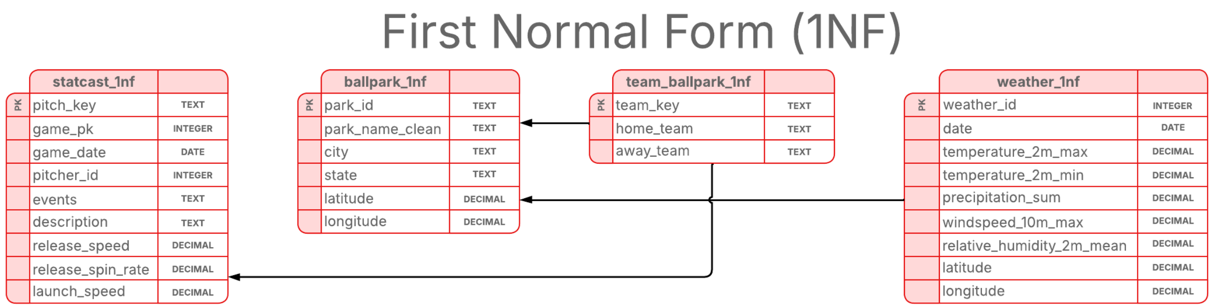
Final ERD:



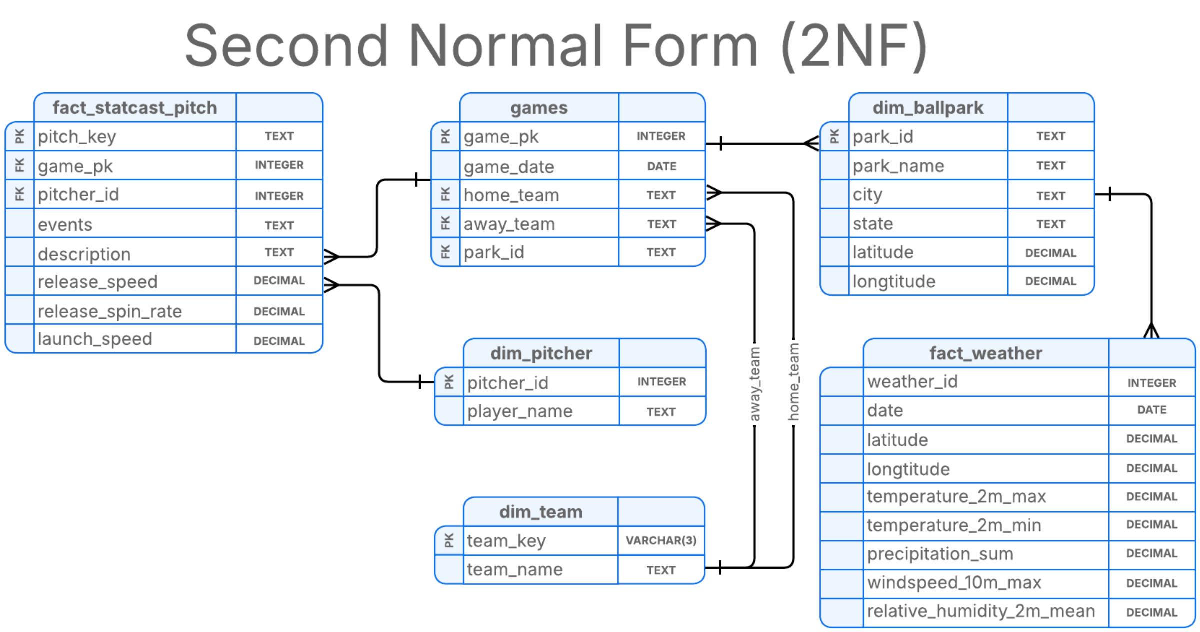
Unnormalized: In the unnormalized stage, the data exists in several separate datasets from different sources, including Seamhead for ballparks, Kaggle for team and stadium relationships, Open-Meteo for weather data, and PyBaseball for pitch and game statistics. At this point, the data is simply collections of attributes without defined relationships or keys. Some information is duplicated across datasets and the relationships between entities like teams, games, ballparks, and weather are not formally structured. This phase helps us identify which attributes exist and how they might relate before organizing them into relational tables.



First Normal Form (1NF): In the first normal form, the data is organized into tables where each column contains atomic values and each row represents a single record. The datasets are separated into tables such as statcast_1nf, ballpark_1nf, team_ballpark_1nf, and weather_1nf, each representing a different entity. This step removes repeating groups and ensures that attributes like pitch metrics, ballpark information, team relationships, and weather measurements are stored in structured tables. While the data is now relational, some attributes still depend on other non-key attributes, which is addressed in the next stage.



Second Normal Form (2NF): In the second normal form, the schema is reorganized to remove partial dependencies so that all attributes depend entirely on their primary key. Pitch-level data is stored in fact_statcast_pitch, while game information is moved to a games table. Player and team data are separated into dim_pitcher and dim_team, and ballpark information is stored in dim_ballpark. Weather data is stored in fact_weather at the grain of one park per day, using park_id and date instead of latitude and longitude to avoid redundancy and ensure consistency with the ballpark dimension. This reduces redundancy and keeps descriptive data stored only once.



Third Normal Form (3NF): In the third normal form, transitive dependencies are removed so that non-key attributes depend only on the primary key. City and state information are separated into a dim_city table, since the city determines the state. The dim_ballpark table references this location information through a foreign key instead of storing the city-state relationship directly. This ensures that each table contains only attributes that depend on its primary key, improving data consistency and reducing duplication.